

3(a)(i)	rate of change of momentum	B1
3(a)(ii)	change in momentum = $(1.4 - 0.80) \times 3.0$	C1
	= 1.8 kg m s^{-1}	A1
3(b)(i)	resultant force (on block) is zero	B1
	(so) velocity is constant	B1
3(b)(ii)	$P = Fv$ or $P = Fs / t$	C1
	$v = 2.0 / 0.80$ ($= 2.5 \text{ m s}^{-1}$)	C1
	distance = 2.5×3.0 = 7.5 m	A1
	or	
	$P = W / t$ or $P = Fs / t$	(C1)
	$W = 2.0 \times 3.0$ ($= 6.0 \text{ J}$)	(C1)
	distance = $6.0 / 0.80$ = 7.5 m	(A1)
3(c)	0 to 3.0 s: upward sloping straight line from the origin.	B1
	3.0 to 6.0 s: horizontal line at non-zero value of momentum with no 'step change' in momentum at 3.0 s	B1